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1. Python Data Wrangling: Failures and Breakthroughs

Challenges: I had been using google collab since a while but trying to run my Jupyter Notebook in Vs Code the Kernel got stuck in a continuous running loop which was happening due to a `ModuleNotFoundError` for pandas.

Solution: I set up a virtual environment, used pip to install the libraries and ran my commands. Also another problem arised where throughout the week the internet bandwidth was really low so internet access was almost 0.

Data Breakthrough: The blood pressure column in the clinical dataset was formatted as a string which couldn't be used for statistical analysis so my breakthrough was to use the `.str.split()` function which is in pandas to separate it into two highly valuable numerical columns as `Systolic_BP` and `Diastolic_BP`.

Data Insights: I noted that there were zero missing values across the three CSV files and during the exploratory data analysis I discovered a moderately positive correlation between age and smoking which is 0.39.

2. SQL Fundamentals : Errors and solutions

The Challenge: I forgot my `SQLWorkBenchPassword`, which I had to crack. The rest was just a revisit because I had already studied SQL Previously so I was used to working with different calculations.

The solution: I did find some new window functions. Some I had not practiced before. Around 10 questions were kind of a new logic for me so I had fun learning something new with that.

AI Utilization Log

I'd consider myself thankful for the AI Advancements in this age since I did utilize it to troubleshoot my problems I faced during the jupyter notebook problem for which I attached screenshots and the llm studied the image and responded carefully. I also used it to revise my SQL knowledge and help me with a couple of questions in a learning manner. I had only generated one chart before, generating multiple side-to-side was also new which I learnt through leverage.